BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 097 · Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does BRA do in LMC?

Q2. When does BRZ branch?

Q3. What does LDA COUNT do?

Q4. Stack Push2,Push5,Pop,Push8: final bottom-to-top?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

termination — Stop condition for loop.

New learning

Observe — BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

Apply — The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

Evaluate — Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Worked example

Start COUNT=2 instead of3. What gets printed and why?

1. Follow the reasoning — Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

We Do · Practise with guidance

In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.

Follow the reasoning

Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

Exit Ticket · Answer before checking

BRZ END always jumps to END regardless of ACC.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Branches unconditionally to a label.

Q2. When accumulator equals zero.

Q3. Loads COUNT's mailbox contents into ACC.

Q4. [2,8].

You Do pages · Independent questions

Q1. Outputs3,2,1 [3]; COUNT values after decrement2,1,0 [3].

Q2. Outputs2,1 [2], COUNT becomes0 [2], BRZ exits before OUT [2].

Q3. BRA always branches [2], so no values printed [2]; BRZ tests ACC zero to allow nonzero iterations [2].

Exit Ticket · Model answer

False. It jumps only when ACC is zero.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: